Lab 2 Deliverable

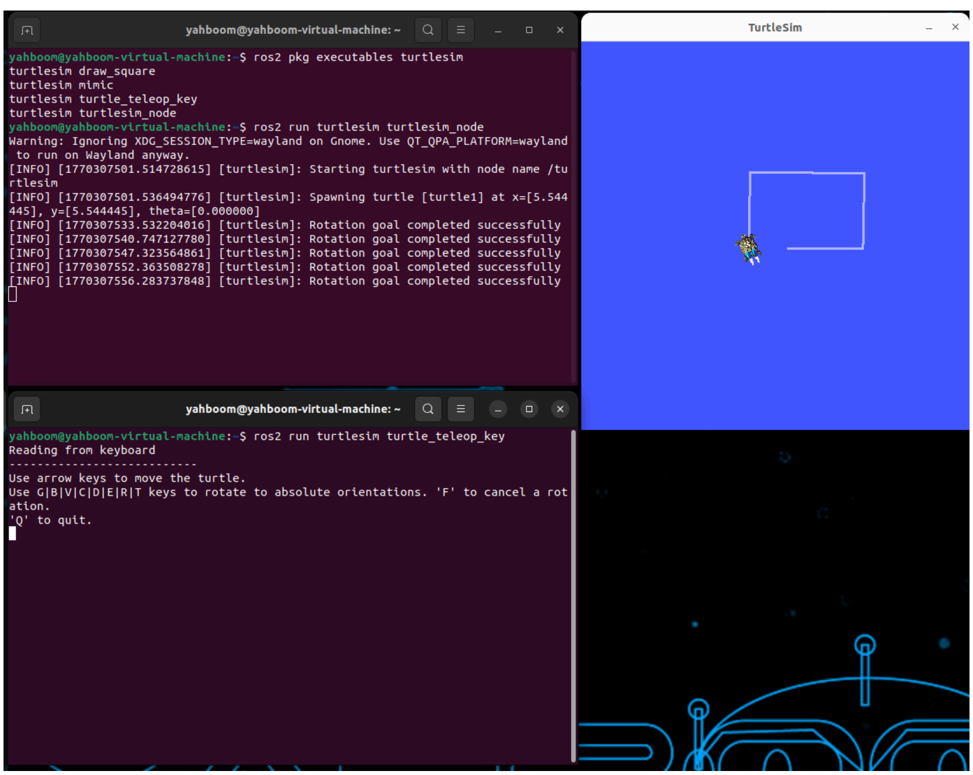
Individual completion proofs should be provided in the Github repository.

- Proof should be clear by adding the instructor to the repository with all members folders containing the appropriate files. Note the commits show who made them so we should see that everyone submitted their own changes.
- For starting nodes: Provide a screenshot showing the multiple terminals.
- For topics: Provide screenshots for steps 7, 21, and 25.
- For services: Provide screenshot for the Challenge.
- For action: Execute step 8 with a unique goal position.
- For task 3: Provide a final image and a text file with a brief explanation of the method you used.

Please label your files as Task#_ followed by the subsection if applicable _step#

Task 2

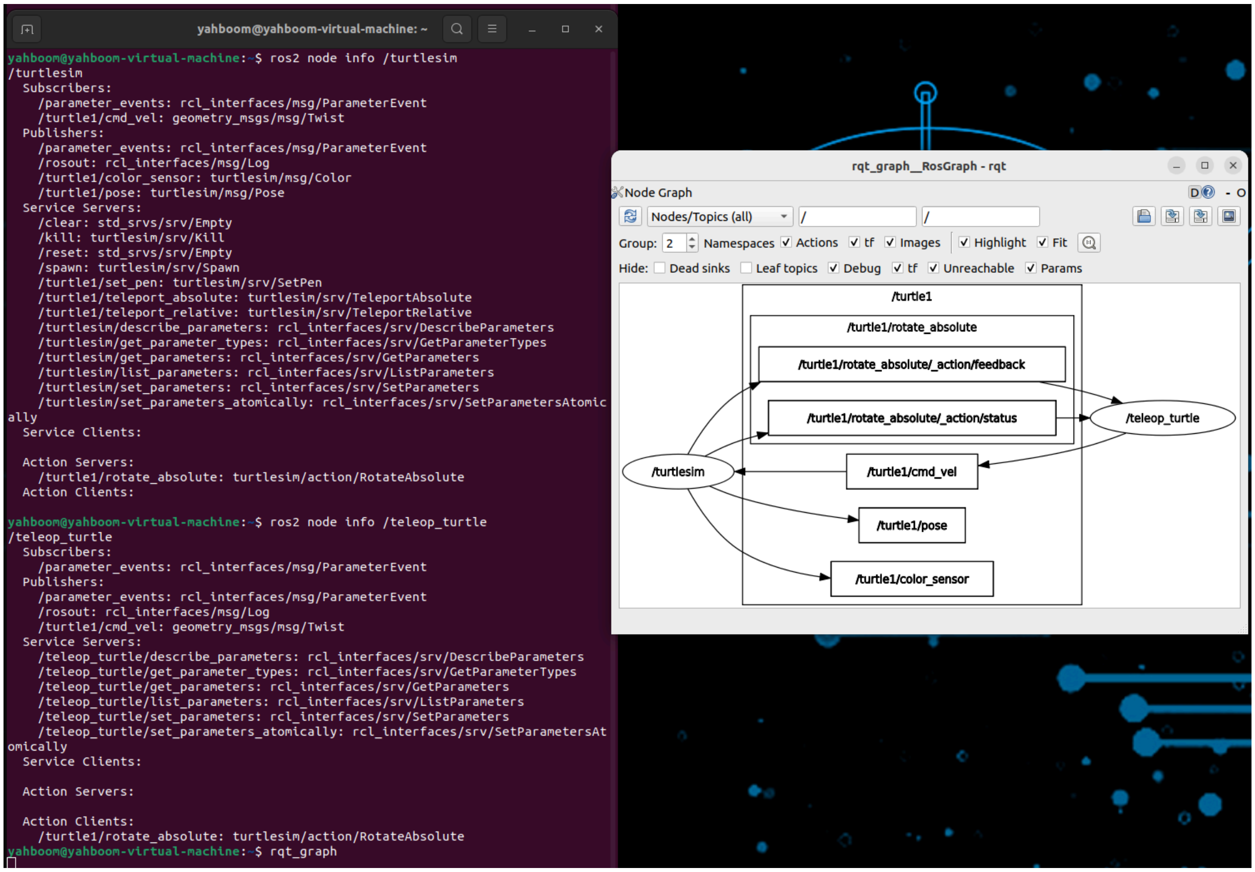
Subtask 1) Starting Nodes



Multiple terminals showing executable nodes for turtlesim package, simulation, and teleop terminals

Subtask 2) Topics

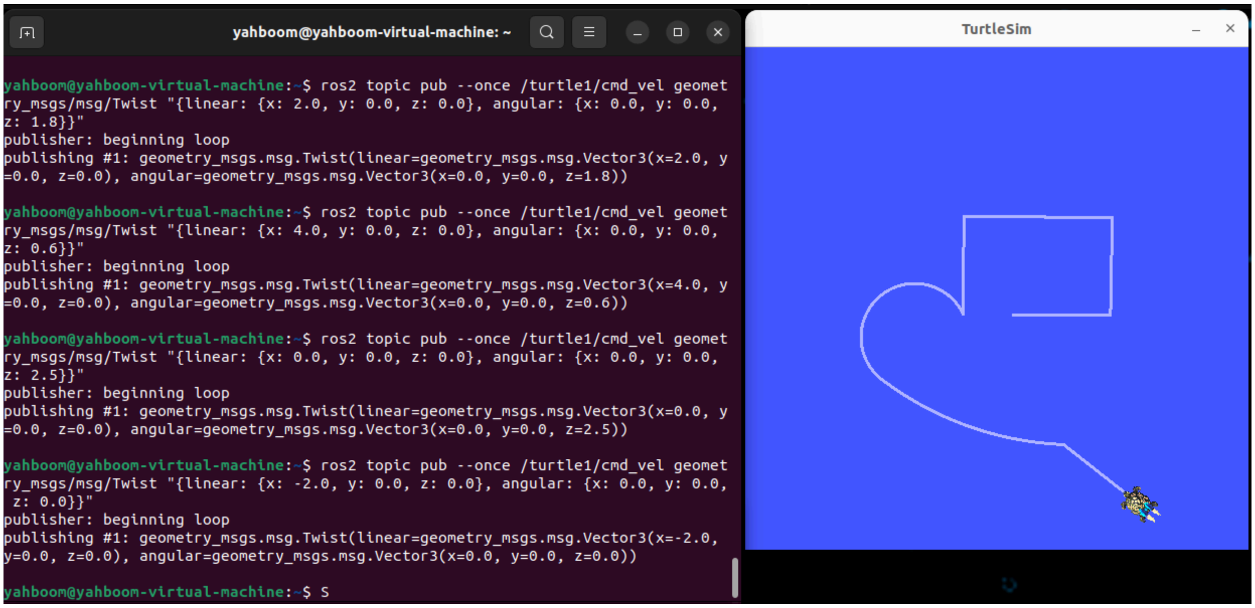
Step #7



Step #21

```
yahboom@yahboom-virtual-machine:~$ ros2 topic info /turtle1/pose
Type: turtlesim/msg/Pose
Publisher count: 1
Subscription count: 0
yahboom@yahboom-virtual-machine:~$ ros2 interface show turtlesim/msg/Pose
float32 x
float32 y
float32 theta

float32 linear_velocity
float32 angular_velocity
yahboom@yahboom-virtual-machine:~$ ros2 topic echo /turtle1/pose
x: 4.509471893310547
y: 5.561339855194092
theta: 2.004814624786377
linear_velocity: 0.0
angular_velocity: 0.0
---
x: 4.509471893310547
y: 5.561339855194092
theta: 2.004814624786377
linear_velocity: 0.0
angular_velocity: 0.0
---
x: 4.509471893310547
y: 5.561339855194092
theta: 2.004814624786377
linear_velocity: 0.0
angular_velocity: 0.0
---
x: 4.509471893310547
y: 5.561339855194092
theta: 2.004814624786377
linear_velocity: 0.0
angular_velocity: 0.0
---
x: 4.509471893310547
```

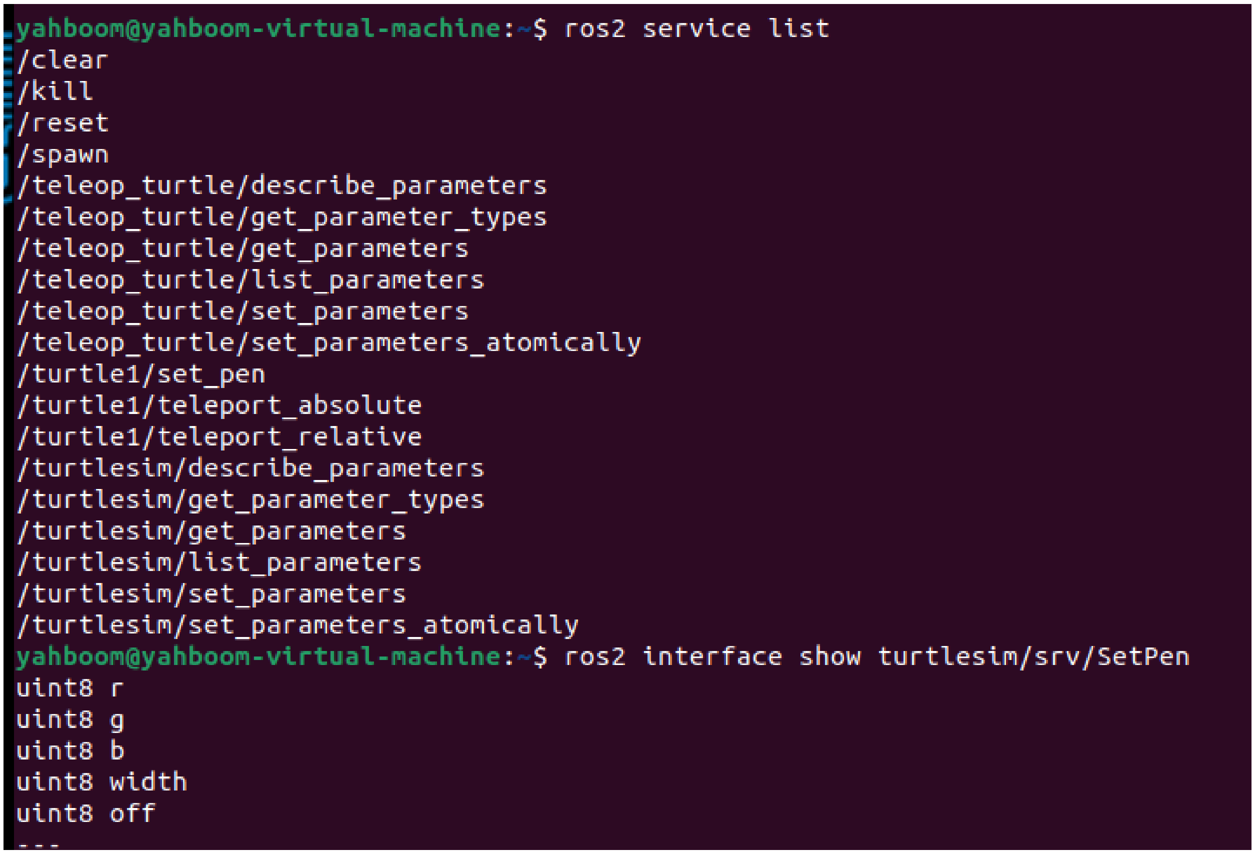
Step #25



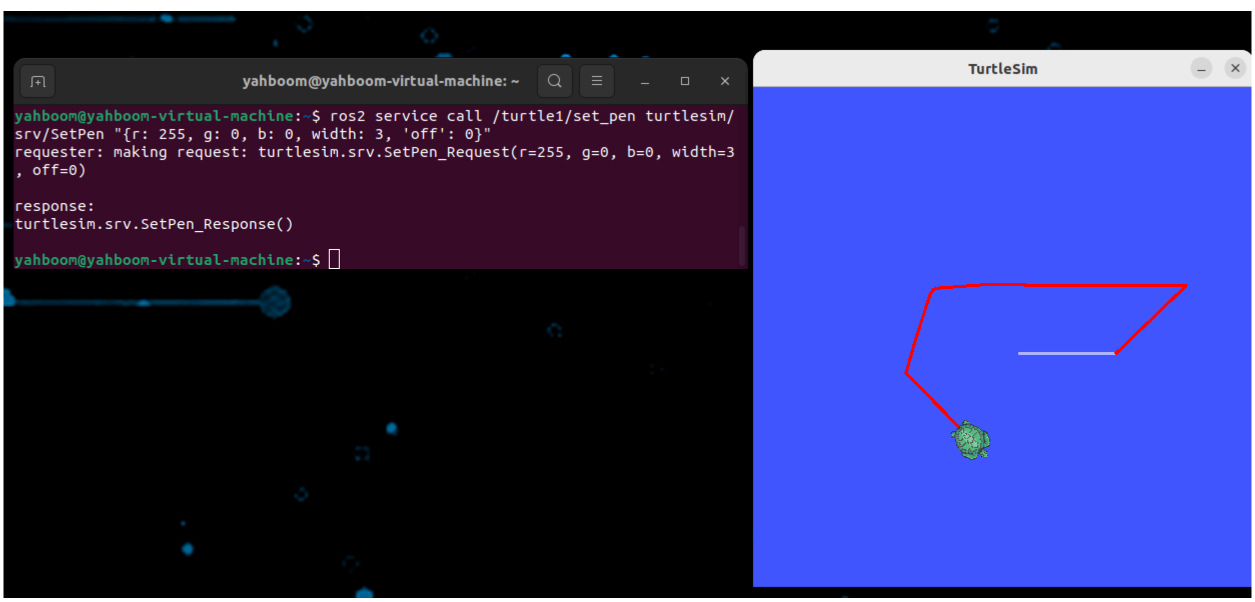
First the lab gives a command with linear x = 2, and angular z = 1.8, this makes the turtle draw a small arc. By making the linear x larger, and angular z smaller. It makes a bigger arc while turning less. Then with only angular values, it makes the turtle spin in place, while negative linear values makes the turtle go backwards.

Subtask 3) Services - Challenge!

The following screenshot shows me finding the service type of /turtle1/set_pen which was found using ros2 service list, and looking at its interface to see how we can modify it.



Then we call the service and changing the interface values to use red pen (r = 255)



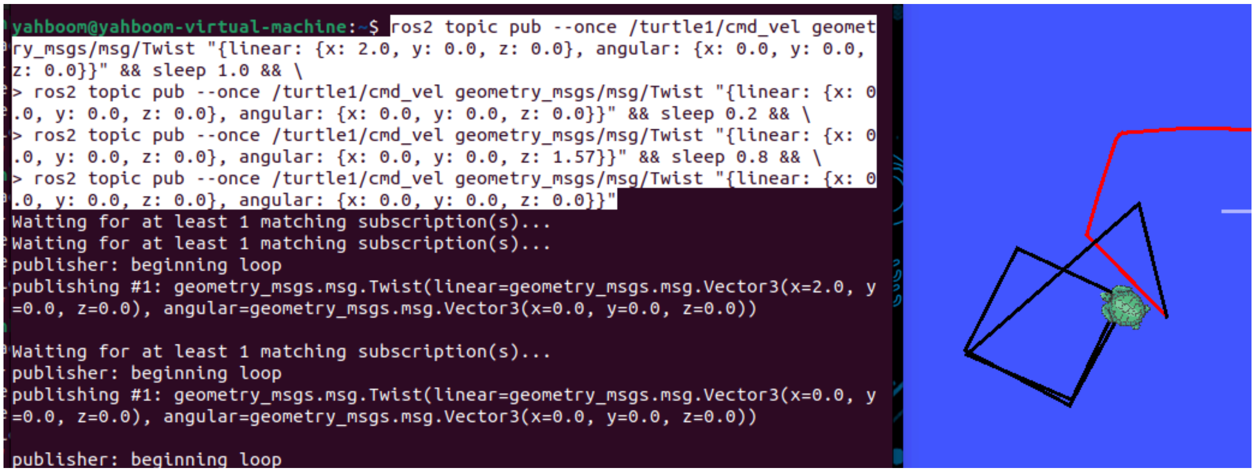
Subtask 3) Actions

Step # 8:

-feedback shows how much theta remaining to be turned until goal theta = 1.8 is met

```
yahboom@yahboom-virtual-machine:~$ ros2 action send_goal /turtle1/rotate_absolu
e turtlesim/action/RotateAbsolute "{theta: 1.8}" --feedback
Waiting for an action server to become available...
Sending goal:
     theta: 1.8

Goal accepted with ID: 6eac00b8b1be4c13900694052c690f85

Feedback:
    remaining: -0.380814790725708

Feedback:
    remaining: -0.36481475830078125

Feedback:
    remaining: -0.3488147258758545

Feedback:
    remaining: -0.33281469345092773

Feedback:
    remaining: -0.316814661026001

Feedback:
    remaining: -0.3008146286010742

Feedback:
    remaining: -0.28481483459472656

Feedback:
    remaining: -0.2688148021697998

Feedback:
    remaining: -0.25281476974487305

Feedback:
    remaining: -0.2368147373199463

Feedback:
    remaining: -0.22081470489501953

Feedback:
    remaining: -0.20481467247009277

Feedback:
    remaining: -0.18881475925445557

Feedback:
    remaining: -0.1728147268295288

Feedback:
    remaining: -0.15681469440460205

Feedback:
    remaining: -0.14081478118896484

Feedback:
    remaining: -0.12481474876403809

Feedback:
    remaining: -0.10881471633911133

Feedback:
    remaining: -0.09281480312347412

Feedback:
    remaining: -0.07681477069854736

Feedback:
    remaining: -0.060814738273620605

Feedback:
    remaining: -0.04481470584869385

Feedback:
    remaining: -0.02881479263305664

Feedback:
    remaining: -0.012814760208129883

Result:
    delta: 0.36800000071525574

Goal finished with status: SUCCEEDED
```

Task 3



First used the command to change pen color to black (r = g = b = 0)

ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 2.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}" && sleep 1.0 && \

ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}" && sleep 0.2 && \

ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 1.57}}" && sleep 1.0 && \

ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}" && sleep 0.2 && \

ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 2.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}" && sleep 1.0 && \

ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}" && sleep 0.2 && \

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ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}" && sleep 0.2 && \

ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 2.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}" && sleep 1.0 && \

ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}" && sleep 0.2 && \

ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 1.57}}" && sleep 1.0 && \

ros2 topic pub --once /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"

I drew the square by publishing velocity commands to /turtle1/cmd_vel using ros2 topic pub --once with geometry_msgs/msg/Twist. Since this topic controls velocity, the distance traveled and the amount of rotation depend on how long each command is applied. I used a forward linear velocity (linear.x = 2.0) and an angular velocity (angular.z = 1.57 = pi/2), along with short sleep delays and stop commands, to control the motion automatically. Repeating this forward-and-turn sequence four times creates four straight sides with 90-degree turns, forming a square.